

DPO

缺陷及其变体

DPOP SimPO LD-DPO IPO ORPO KTO

东川路第一可爱猫猫虫

主要内容

- 缺陷一

偏好回答和不偏好回答的概率可能会一起下降

DPOP

- 缺陷二

length bias

SimPO LD-DPO

- 缺陷三

过拟合

IPO

- ORPO KTO

DPO

- $\text{loss} = -E_{(x, y_w, y_l) \sim \mathbb{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$

$\hat{r}_{\theta}(x, y) = \beta \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)}$ 是由模型的概率比差定义的隐式奖励

ence model π_{ref} (more in Section 5). Intuitively, the gradient of the loss function $\mathcal{L}_{\text{DPO}}$ increases the likelihood of the preferred completions y_w and decreases the likelihood of dispreferred completions y_l . Importantly, the examples are weighed by how much higher the implicit reward model $\hat{r}_{\theta}$ rates

- 理想情况下

我们希望看到

$\pi_{\theta}(y_w | x)$ 升高, $\pi_{\theta}(y_l | x)$ 降低

缺陷一

- 偏好回答和不偏好回答的概率可能会一起下降



- DPO的损失函数是概率比的差值
即使 $\pi_{\theta}(y_w | x)$ 被降低了
只要 $\pi_{\theta}(y_l | x)$ 被降低的更多, loss依然是下降的

1, as long as $\pi_{\text{ratio}}(y_l|x)$ is also lowered sufficiently. This implies that the log-likelihood of the preferred completions is reduced below the original log-likelihood from the reference model.

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Smaug: Fixing Failure Modes of Preference Optimisation with DPO-Positive

Arka Pal, Deep Karkhanis, Samuel Dooley, Manley Roberts, Siddartha Naidu, Colin White

- RLHF 的最初用例并没有明确指出偏好回答 y_w 也就是令人满意的回答
只是相对地优于另一个回答
- 可是后来更常见的是
对于每对回答, 其中一个是令人满意的回答
 $y_w: 2+2=4$ $y_l: 2+2=5$

编辑距离较近的两个回答

- $y_w: 2+2=4$ $y_l: 2+2=5$

编辑距离为1

- 当 y_w 和 y_l 几乎相同时

DPO 的梯度更新会导致模型去抑制这些词的生成

偏好回答和不偏好回答同时被抑制

over another. In this work, first we show theoretically that the standard DPO loss can lead to *a reduction of the model's likelihood of the preferred examples*, as long as the relative probability between the preferred and dispreferred classes increases. We then show empirically that *this phenomenon occurs when fine-tuning LLMs on common datasets, especially datasets in which the edit distance between pairs of completions is low*. Using these insights, we design DPO-Positive (DPOP), a new loss function

3D-PROPERTIES: IDENTIFYING CHALLENGES IN DPO AND CHARTING A PATH FORWARD

Property 1 (Drastic drop in rejected response likelihood). The likelihood of a rejected response tends to shift much more rapidly than that of a chosen response.

Property 2 (Degradation into response suppression). As optimization progresses, DPO gradually loses its ability to steer the direction of optimizing chosen responses and instead devolves into merely suppressing the rejected responses.

Property 3 (Dispersion effect on unseen responses). As DPO training progresses, the likelihood of both chosen and rejected responses gradually decreases, while the likelihood of generating out-of-distribution (OOD) responses increases.

Direct Preference Learning

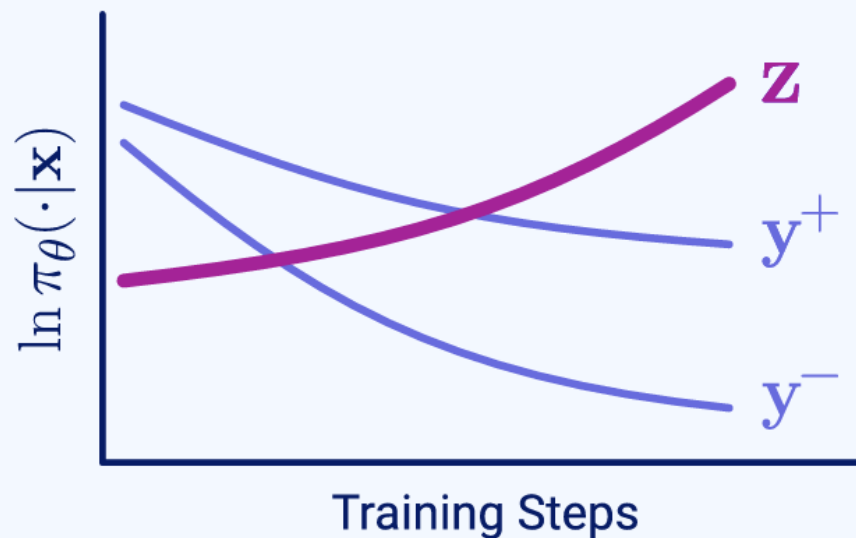
e.g. DPO (Rafailov et al. 2023)

$\equiv \mathbf{x}$ $\equiv y^+$ $\equiv y^-$

Language Model π_θ

Preference Learning Loss

Likelihood Displacement



Benign

Z is as preferable as y^+

Catastrophic

Z is substantially less preferable than y^+

tual) hidden embeddings of y^+ and y^- . In particular, it formalizes intuition by which the more similar y^+ and y^- are the more $\ln \pi_\theta(y^+ | \mathbf{x})$ tends to decrease.

- Likelihood Displacement 概率错位

如果概率质量从偏好回答转移到了具有相反含义的回答上
这是灾难性的

DPOP

The full DPOP loss function is thus:

$$\mathcal{L}_{\text{DPOP}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim D} \left[\log \sigma \left(\beta \left(\log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) - \lambda \cdot \max \left(0, \log \frac{\pi_{\text{ref}}(y_w|x)}{\pi_{\theta}(y_w|x)} \right) \right) \right]$$

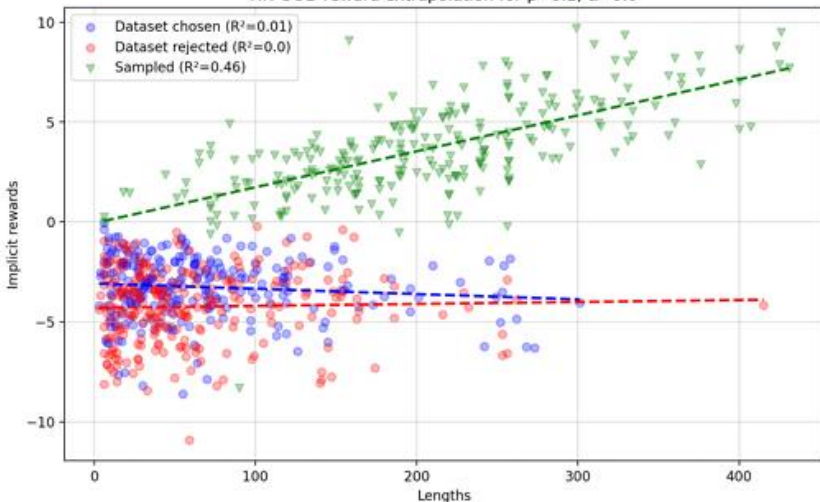
- $\mathcal{L}_{\text{DPOP}} = \mathcal{L}_{\text{DPO}} + \lambda \cdot \max(0, \log \pi_{\text{ref}}(y_w|x) - \log \pi_{\theta}(y_w|x))$
- 当模型生成的正例概率低于参考模型时
新增的惩罚项才起作用
- 模型在训练过程中， 必须努力保持正例的概率不低于初始水平
- Smaug-72B-v0.1

“HuggingFace排行榜上首个平均得分超过80%的开源模型”

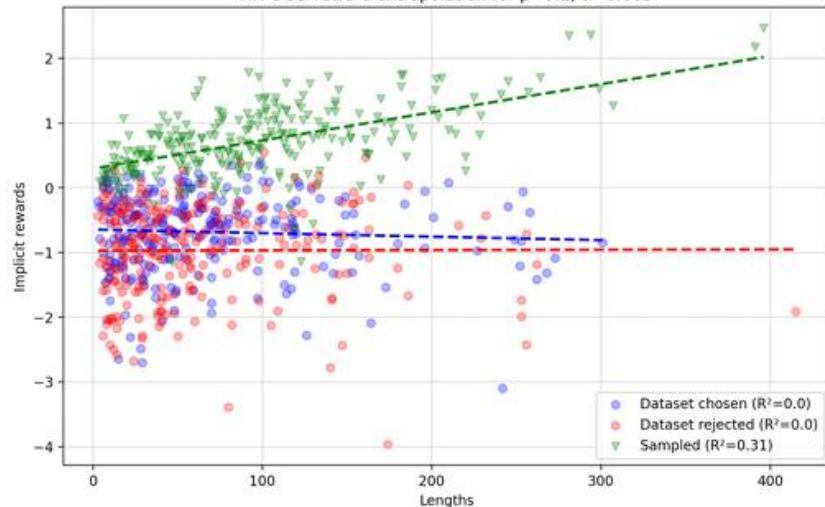
缺陷二 length bias

- 长度偏置 非理性冗长
- 即使是一些简单的问题，DPO后的模型也更喜欢长篇大论
即使在明确简洁的指令下，仍出现可压缩的过长回答
- 两大类原因
 - 第一类来自数据本身
 - 人类通过增加字数来应对误解风险或情感连接需求
 - 第二类原因来自DPO算法
 - 序列级隐式奖励信号随输出长度增长

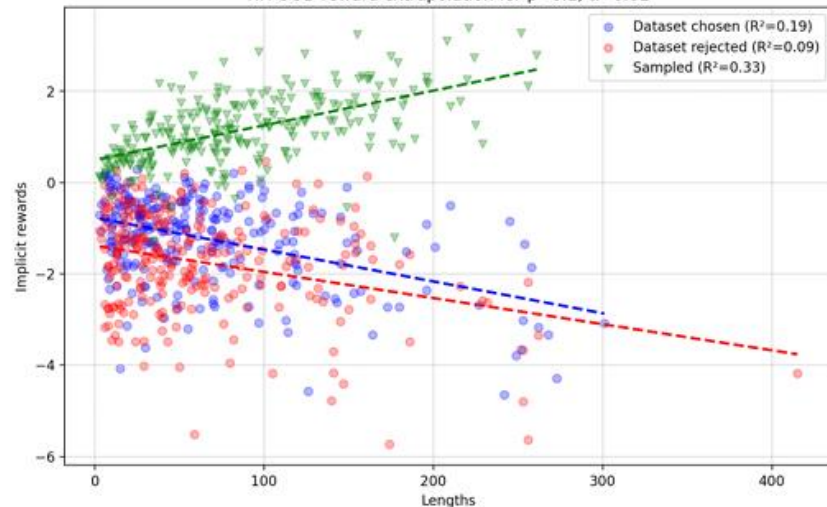
HH OOD reward extrapolation for $\beta=0.1, \alpha=0.0$



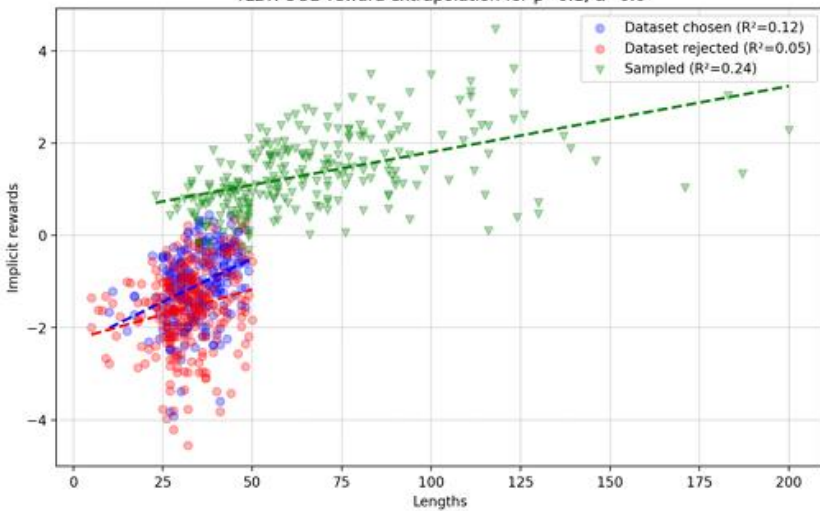
HH OOD reward extrapolation for $\beta=0.1, \alpha=0.005$



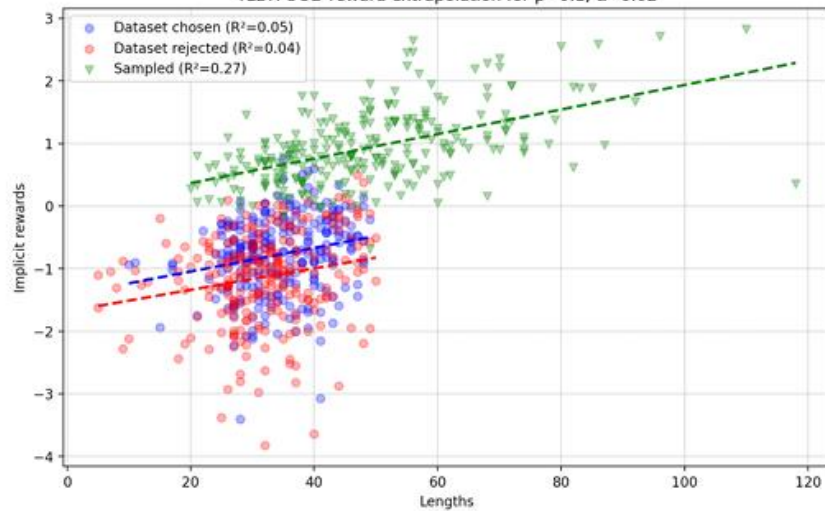
HH OOD reward extrapolation for $\beta=0.1, \alpha=0.01$



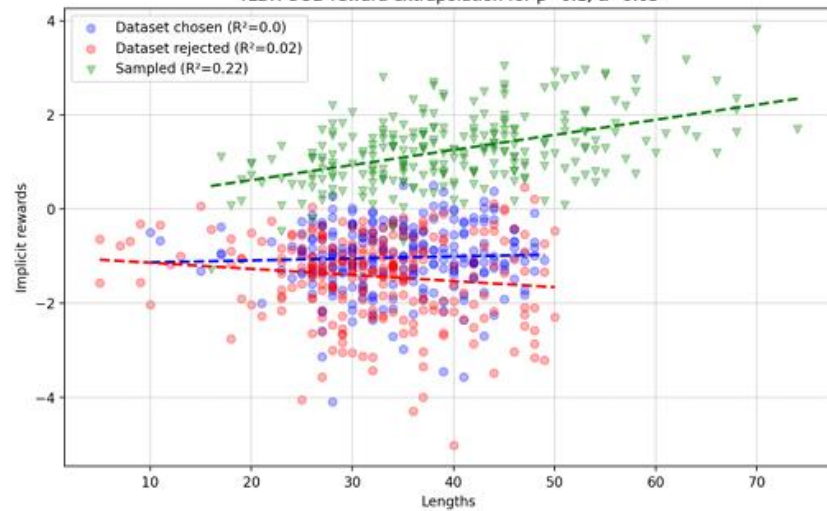
TLDR OOD reward extrapolation for $\beta=0.1, \alpha=0.0$



TLDR OOD reward extrapolation for $\beta=0.1, \alpha=0.02$



TLDR OOD reward extrapolation for $\beta=0.1, \alpha=0.05$



DPO算法导致length bias

- $r(x, y) = \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)}$

- 这个序列概率是每个token概率的连乘

$$\pi_{\theta}(y|x) = \prod_{t=1}^{\text{len}(y)} p(y_t|x, y_{<t})$$

$$r(x, y) = \beta \sum_{t=1}^{|y|} \log \frac{p_{\theta}(y_t|x, y_{<t})}{\pi_{\text{ref}}(y_t|x, y_{<t})}$$

- DPO的训练目标是赢的序列奖励总和最大
这就导致一种 reward hacking

SimPO

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) =$$

$$-\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right]$$

$$\mathcal{L}_{\text{SimPO}}(\pi_{\theta}) =$$

$$-\mathbb{E} \left[\log \sigma \left(\frac{\beta}{|y_w|} \log \pi_{\theta}(y_w | x) - \frac{\beta}{|y_l|} \log \pi_{\theta}(y_l | x) - \gamma \right) \right]$$

- 长度归一化
- 移除参考模型
- $\sigma(r(x, y_w) - r(x, y_l) - \gamma)$

LD-DPO

- 做法：降低超长部分的权重
- 对于一对偏好回答 (l_w, l_l)

$$\text{令 } l_p = \min(l_w, l_l)$$

- 将公共部分与超长部分拆分

$$\pi_{\theta}(y|x) = \prod_{i=1}^{l_p} p(y_i|x, y_{<i}) \cdot \prod_{i=l_p+1}^l p(y_i|x, y_{<i})$$

- 修改超长部分的似然函数

$$\widehat{\pi}_{\theta}(y|x) = \prod_{i=1}^{l_p} p(y_i|x, y_{<i}) \cdot \prod_{i=l_p+1}^l p(y_i|x, y_{<i})^{\alpha}$$

$$\log \widehat{\pi}_{\theta}(y|x) = \sum_{i=1}^{l_p} \log p_i + \alpha \sum_{i=l_p+1}^l \log p_i$$

缺陷三 过拟合

- $h_{\pi}(y, y', x) = [\log \pi(y|x) - \log \pi(y'|x)] - [\log \pi_{\text{ref}}(y|x) - \log \pi_{\text{ref}}(y'|x)]$
- $\text{loss} = -\log \sigma(\beta h_{\pi})$
- $\frac{\pi(y|x)}{\pi(y'|x)} = \exp(h_{\pi}(y, y', x)) \cdot \frac{\pi_{\text{ref}}(y|x)}{\pi_{\text{ref}}(y'|x)}$
- Identity Preference Optimization (ΨPO with identity mapping)
- $E_{(y_w, y_l, x) \sim D} \left(h_{\pi}(y_w, y_l, x) - \frac{1}{2\tau} \right)^2$
目标间隔: $\frac{1}{2\tau}$

ORPO

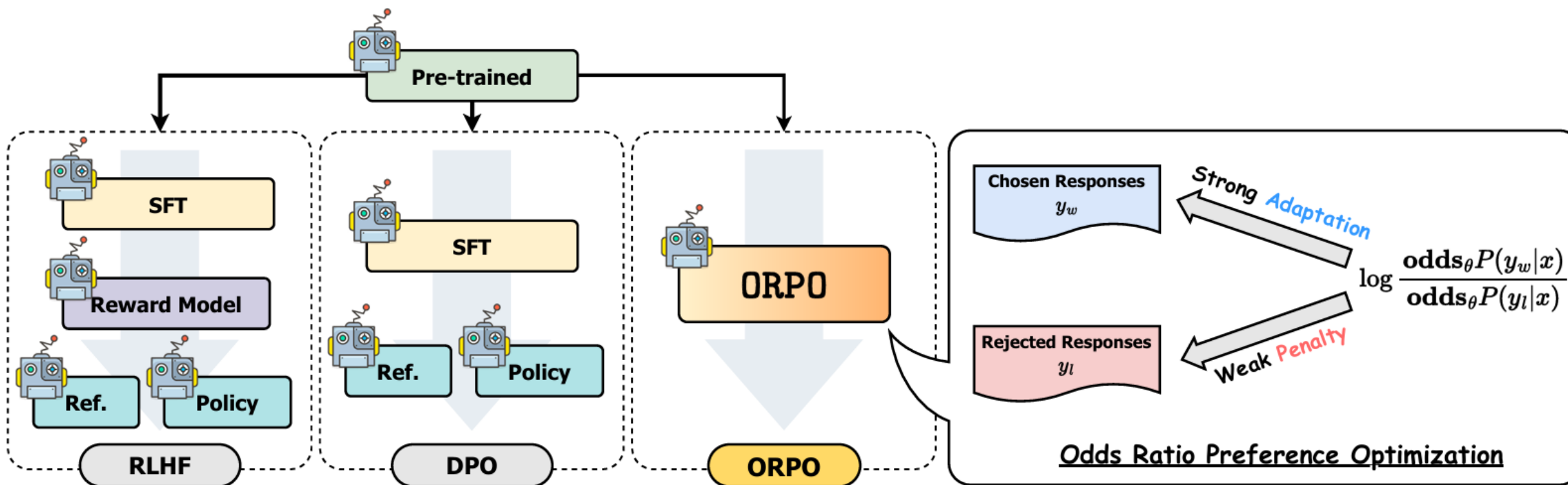
ORPO: Monolithic Preference Optimization without Reference Model

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odds ratio preference optimization



- 不需要SFT: Monolithic reference-free



- SFT会同时让chosen和rejected的概率提高
- 传统SFT无法区分chosen和rejected
所以才需要RLHF DPO
- ORPO希望在一个阶段里
同时完成过去SFT和DPO两个阶段做的事情

ORPO的损失函数

- $\mathcal{L}_{ORPO} = E_{(x, y_w, y_l)} [\mathcal{L}_{SFT} + \lambda \cdot \mathcal{L}_{OR}]$

- 胜算定义为:

$$odds_{\theta}(y|x) = \frac{P_{\theta}(y|x)}{1-P_{\theta}(y|x)}$$

$$\log P_{\theta}(y|x) = \frac{1}{m} \sum_{t=1}^m \log P_{\theta}(y_t|x, y_{<t})$$

- 胜算损失 $\mathcal{L}_{OR}$:

$$\mathcal{L}_{OR} = -\log \sigma \left(\log \frac{odds_{\theta}(y_w|x)}{odds_{\theta}(y_l|x)} \right)$$

当偏好响应的胜算远大于不偏好响应时, 该损失趋于最小

KTO

KTO: Model Alignment as Prospect Theoretic Optimization

Kawin Ethayarajh¹ Winnie Xu² Niklas Muennighoff² Dan Jurafsky¹ Douwe Kiela^{1 2}

KTO不再需要偏好数据



- 只需要一个二值标签表示回答好还是不好
这样的数据更易得到
- 如果我拥有的是DPO的偏好数据(y_w, y_l, x)
只需拆分开
用同一组数据可以清晰比较DPO和KTO性能
- 如果我拥有的是打分数据，如何转换来做DPO？
只要得分高于某个阈值，就是好的
只要得分高于均值或中位数，就是好的
按分数映射成伯努利分布再抽样

- $r_{\text{KTO}}(x, y) = \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)}$

- 参考点:

$$z_{\text{ref}} = \mathbb{E}_{\mathbf{x}' \sim \mathbb{D}} \left[\beta \text{KL} \left(\pi_{\theta}(\cdot | x') || \pi_{\text{ref}}(\cdot | x') \right) \right]$$

- 相对奖励:

$$z = r_{\text{KTO}} - z_{\text{ref}}$$

$$\text{desirable: } v_{\text{KTO}} = \sigma(r_{\text{KTO}} - z_{\text{ref}})$$

$$\text{undesirable: } v_{\text{KTO}} = \sigma(z_{\text{ref}} - r_{\text{KTO}})$$

- KTO的损失函数

$$L_{\text{KTO}}(\pi_{\theta}, \pi_{\text{ref}}) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{D}} \left[w(y) \left(1 - v_{\text{KTO}}(x, y; \beta) \right) \right]$$

样本权重 $w(y)$: desirable用 λ_D , undesirable用 λ_U

$$\nabla_{\theta} L_{\text{KTO}}(\pi_{\theta}, \pi_{\text{ref}}) = \mathbb{E}_{x, y \sim D} [\lambda(y) \sigma(z) \sigma(-z) \nabla \beta \log \pi_{\theta}(y|x)]$$

- 梯度里含有 $\sigma(z)\sigma(-z)$

当 z 很大或者很小的时候梯度趋于0

KTO不会从极高reward的坏样本/极低reward的好样本里学习

- KTO是以全局reference为中心的两侧推拉

desirable: $\lambda(y) = -\lambda_D < 0$

梯度下降等价于上升 $\log \pi_{\theta}(y|x)$

undesirable: $\lambda(y) = \lambda_U > 0$

梯度下降等价于下降 $\log \pi_{\theta}(y|x)$

- KTO比DPO更免疫整体的概率偏移

KTO的优势

- 不需要成对偏好

比如评论区只有点赞或点踩的业务场景下
点赞点踩这样的二值反馈更丰富、更便宜、也更容易快速收集
极端数据场景下KTO还能工作

好坏配比1:10 一样可以工作

one-y-per-x : (y_w, y_l, x) 里的每个x随机保留一个y

- KTO 对噪声和极端或矛盾样本更鲁棒

不学习标签错了但模型很自信的样本，不学习极端 outlier

- 不做 SFT 时，KTO 明显更稳定